INFINITE / AUDIOVISUAL NODE WORKSTATION

The Headless *Engine*

Render pictures, video and sound from a plain text file. No window, no mouse, same Infinite underneath. A friendly guide to the command line, the patch format, and the loop that gets you from idea to picture.

Runs on macOS, Windows, Linux

Version 0.6.0

Output PNG, MP4, MOV, WAV

FIVE CABLES, FIVE JOBS

The same cable colours you see in the app, now written as one line each in a patch file.

- `cable` Image
- `mod` Modulator
- `aud` Audio
- `note` Notes
- `pal` Palette

BEFORE YOU START

What is the headless engine?

Render pictures, video and sound from a text file, with no window and no mouse. The headless engine is the same Infinite you already have, driven from the command line. Anything a patch can do in the app (3D, Field kernels, modulation, synths, effects) works here, and the output is deterministic: the same patch gives the same frames and the same samples.

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SECTION 01

What you need

NEED	DETAIL
The Infinite app	There is no separate package. The executable inside the app is the command-line tool.
A graphics driver	Frames are drawn by the real GPU pipeline, so a machine with working OpenGL is required.
Linux only: a virtual display	Run every render as `xvfb-run -a Infinite ...`. Without it you get `E_NO_DISPLAY`.
A text editor	Patches are plain text files ending in `.inf`.

You do not need: a project file, the app open, a licence dialog, or any knowledge of the node graph beyond the nodes you use.

Not supported headless: patches that use a camera, microphone or MIDI device are refused (exit code 4), because there is no hardware to read. Replace them with a file or a generator.

The executable

PLATFORM	PATH
macOS	`Infinite.app/Contents/MacOS/Infinite`
Windows	`Infinite.exe`
Linux	`Infinite`

The examples below write it as `Infinite`. On macOS, make it a shortcut once:

```
alias Infinite="/Applications/Infinite.app/Contents/MacOS/Infinite"
```

Check it works: `Infinite --version` prints the version number.

SECTION 02

Your first render in five minutes

Save this as `hello.inf`:

```
infinite-patch 1
node 1 Source Shape
  id shape
  i shapeType Star
end
node 2 Modulators LFO
  id lfo
end
node 3 Utility Output
  id out
end
cable out 0 shape
mod shape sizeX lfo 0 0 1 0.5
expr shape rotation sin(t) * 90
```

Then run these in order:

```
Infinite --validate hello.inf
Infinite --frame hello.inf 0,1,2 pics/ --contact-sheet sheet.png
Infinite --render hello.inf hello.mp4 --duration 4 --fps 30
```

STEP	WHAT IT DOES
`--validate`	Loads the file strictly and lists every problem with its line number. Exit code 0 means it is good.
`--frame`	Draws stills at 0, 1 and 2 seconds, and one contact sheet with each cell labelled by time.
`--render`	Writes a video.

A star pulses and spins. That is a complete patch: one shape, one modulator, one output.

SECTION 03

The working loop

Always work in this order. Do not guess a node's name or a parameter's range; ask the engine.

1. **Describe.** `Infinite --describe "<node type>"` lists the node's controls (key, default, min, max, dropdown options), inputs and outputs. `Infinite --describe` alone lists every node type. Add `--json file.json` for a machine-readable version.
2. **Write** the smallest patch that could work. Leave every control you do not change at its default.
3. **Validate.** `Infinite --validate patch.inf` . Fix errors until it exits 0. Each error names the line and suggests the nearest valid name.
4. **Explain.** `Infinite --explain patch.inf` . It prints the graph that was actually built. Check that every wire you wrote is there and that nothing important is still at a default you meant to change.
5. **Look and listen.** `--frame` for pictures, `--audio-summary` for sound.
6. **Fix** one thing at a time, and go back to step 3.
7. **Render** the final file.

A patch can validate and still be the wrong patch, which is why step 4 exists.

SECTION 04

Command reference

4.1 Commands

COMMAND	PURPOSE
`--describe [type] [--json file]`	List node types, or one type's controls, inputs and outputs
`--validate patch.inf [--for-render] [--lenient] [--json file]`	Check the file loads; report every problem
`--explain patch.inf [--json] [--all] [--lenient]`	Print the built graph: nodes, non-default controls, wires, modulation ranges, world bounding boxes of geometry. `--all` lists every control
`--frame patch.inf T[,T...] out.png or dir/`	Still images at the given times in seconds. Several times need a folder
`--frames-dir patch.inf out_dir --duration S`	A numbered PNG sequence. Options: `--start S` , `--fps N` , `--alpha` , `--depth 8 or 16`

`--render patch.inf out.mp4 or out.mov`	A video with audio
`--audio-summary patch.inf out.json`	The sound as numbers: loudness, peaks, clipping, silence, spectrum, tempo guess
`--canonicalize in.inf out.inf [--keep-ids]`	Rewrite a patch in the numeric form the app saves
`--version`	Print the version

4.2 Options

OPTION	APPLIES TO	MEANING
`--start S`	render, frames-dir, audio-summary	First second to render (default 0)
`--duration S`	render, frames-dir, audio-summary	Length in seconds. Default: the Output node's own length, else 10
`--fps N`	render, frames-dir	Frames per second, 1 to 240
`--size WxH`	frame, frames-dir, render	Override the output size, 1x1 up to 16384x16384
`--bpm N`	render, audio-summary	Tempo, 1 to 999
`--sample-rate N`	audio	8000 to 384000
`--codec h264 or prores4444`	render	Video codec. Use `prores4444` (in a `.mov`) when you need transparency
`--no-audio`	render	Write a silent video
`--node index[:output]`	frame, frames-dir, render	Render this node instead of the Output. Great for checking one stage
`--set node.param=value`	frame, frames-dir, render	Override one control for this run without editing the file
`--output node`	audio-summary	Measure one Audio Out or Output instead of the whole mix
`--wav out.wav`	audio-summary	Also write the audio as a WAV
`--stems id,id --stems-dir dir`	audio-summary	Write one WAV per named node
`--notes events.json --note-map map.json`	audio-summary	Play your own note events into the patch
`--timeline`	audio-summary	Play the arrangement's audio lanes instead of the node graph alone
`--plugin-params`	render, audio-summary	List every loaded plugin's parameters (address, name, real range, default) in the `plugins` block of the result
`--contact-sheet sheet.png`	frame	One image holding every requested frame
`--png-level 0-9`	frame, frames-dir	PNG compression
`--lenient`	all	Treat unknown keys and removed nodes as warnings, not errors
`--json file`	describe, validate, canonicalize	Write the result as JSON to a file
`--timeout S`	all	Stop with exit code 6 after this many seconds

4.3 Reading the result

The **last line printed** is always one JSON object with these fields: `ok`, `mode`, `frames`, `fps`, `size`, `audio`, `elapsed_ms`, `files`, `warnings`, `errors`. Renders of a patch with plugins add a `plugins` block saying which loaded and which did not. Frame renders add `frame_stats` (brightness, black and white percentages, a histogram per frame). Audio summaries add `audio_summary`. Each issue is `{code, message, line, node, hint}`.

4.4 Exit codes

EXIT	MEANING
0	Success
2	Bad command line
3	The patch does not load or validate. Read `errors[0]`
4	Refused: the patch needs a camera, microphone or MIDI device
5	Rendering failed
6	Timeout. Raise `--timeout`

SECTION 05

The patch file format

A patch is a line-based text file. The first line is `infinite-patch 1`. Blank lines and lines starting with `#` are ignored.

```
node <index> <category> <type name>
  id <word>                optional name you can use instead of the index
  f <key> <float>          a number
  i <key> <int or option>  a whole number or dropdown choice (i shapeType Star)
  b <key> <0 or 1>         on/off
  c <key> <r> <g> <b>      a colour, each 0 to 1
  s <key> <text>           text, to the end of the line
end
cable <dst> <slot> <src>   image wire
geo   <dst> <slot> <src>   geometry, camera or light wire
aud   <dst> <slot> <src>   audio wire
note  <dst> <slot> <src>   note wire
mod   <dst> <key> <src> <srcOutput> <polarity> <depth> <centre> [<lo> <hi>]
expr  <dst> <key> <expression>
glob  <name> <expression>
```

Rules worth remembering:

- **Wires read destination first.** `cable out 0 shape` means "Output slot 0 is fed by shape".
- **Names work everywhere.** Use a node's `id` instead of its number, a slot's name (`input`, `input_2`) instead of its position, and a control's key in `mod` and `expr`.
- **Leave defaults out.** A node needs only the keys it changes.
- **Free text goes last** on its line.
- **No sigils in expressions.** Write `P.y += bass * 2`, never `@P.y`.
- **Bypass.** A node with two or more inputs never bypasses, and a bypassed node does not cook.

- **Something must be reachable.** Pictures need a `Utility Output`. Sound needs a `Utility Audio Out`, or an Output with audio wired in.
- **Animate inside the patch.** Use modulators, `mod`, `expr` and `glob`. Do not render one frame at a time with edited values.

The full record reference is `docs/reference/patch-format.md`. Three small working patches (picture, sound, modulation) are in `assets/examples/authoring/`.

SECTION 06

Modulation: making things move

Three ways, from simplest to most flexible.

WAY	EXAMPLE	USE IT FOR
`expr`	`expr shape rotation sin(t) * 90`	A formula of time `t`
`mod`	`mod shape sizeX lfo 0 0 1 0.5`	A modulator node (LFO, Random, Envelope, Audio Analyze) driving a control
`glob`	`glob wobble sin(t * 2)`	A named value many controls can share

Range of a `mod`. The two optional numbers after the centre, `lo` and `hi`, are the real values the control sweeps between. They are absolute, not percentages. If a sphere's scale is 0.5 and you want it to breathe by 40%, write `0.4` and `0.7`, not `0.8` and `1.4`.

Which controls can be modulated. Only some controls accept a modulator. `Infinite --describe "<type>" --json` marks them with a `modulatable_index`. If a control has none, it cannot be driven; use a node that does (for example a `Transform` placed after a 3D primitive).

Sound driving picture. Wire a final audio node into an `Audio Analyze` node. It outputs level, low, mid, high, onset and eight frequency bands, each of which can drive any modulatable control. Chain order matters: analyse the signal after your effects so the picture follows what you hear.

SECTION 07

Working with 3D

3D follows the same loop, with a few rules that save time.

- Primitives (Sphere, Cone, Cylinder, Torus, …) have position, scale and rotation, but those are fixed, not modulatable. To animate a shape, send it through a **Transform** node and modulate that.
- A Transform applies scale and rotation about the shape's own centre, then moves it by its offset. Create the primitive at the origin at size 1, and put its size, rotation and position on the Transform. If a primitive is already scaled or moved, the Transform's offset will not land where you expect.
- Join shapes with **Join Geometry**, give them a **Material**, and send the result to **Render 3D** together with a **Camera** and one or more **Lights**.
- `Infinite --explain patch.inf --all` prints each piece's world bounding box. Use it to check where things really are before blaming the lights.
- If the picture is black or faint, check the camera distance, the light intensities, and the Render 3D ambient colour before changing anything else.

SECTION 08

Troubleshooting

SYMPTOM	WHAT TO CHECK

Black frame, `W_BLACK_FRAME`	Is the Output wired (`W_OUTPUT_EMPTY`)? Is a source's size or colour zero? Is a node upstream bypassed?
Silent audio, `W_SILENT`	Is an Audio Out reachable from a synth? Do notes reach the synth? Is a volume at 0?
`W_UNUSED_NODE`	The node reaches no Output or Audio Out. Wire it in or delete it
`E_FIELD_COMPILE`	A Field node's code does not compile. `--validate` compiles Formula and Equation formulas, expressions and globs too, so you find out before rendering. The message gives the line
`W_PLUGIN_NOT_LOADED`	A plugin in the patch could not be loaded (missing, or it crashed the app before). The node passes the sound through. Check `plugins` in the result
`W_NOTE_FANOUT`	One note output feeds more than 64 consumers; the extra ones get no notes
`W_CLIPPING`	Samples hit full scale. Lower a gain, or add a Limiter
`E_KIND_MISMATCH`	You wired the wrong kind of signal (image into an audio slot). `--describe` lists each slot's kind
`E_BAD_KEY` or `E_BAD_VALUE`	Misspelled control or dropdown option. The hint names the nearest valid one
`E_HARDWARE_SOURCE` (exit 4)	The patch has a camera, microphone or MIDI source. Replace it
`E_NO_DISPLAY` (Linux)	Run under `xvfb-run -a`
It validates but looks wrong	Run `--explain` and compare to what you meant
Something moves too far or too little	A `mod` range is absolute. Check `lo` and `hi`

SECTION 09

Good practice

- Keep patches in a folder under version control. They are plain text and diff cleanly.
- Render a contact sheet before a full video. A 4-frame sheet costs seconds.
- Use `--set` to try a value without editing the file.
- Use `--node` to look at one stage of a long chain.
- Quote the command line when a patch lives in a folder with spaces.
- The output is deterministic, so a patch that renders correctly once renders the same way again. Frame renders hold the clock at zero from the first tick, so history nodes such as Time Machine render the same every run, and headless runs never read or write the learned prediction profiles. One known exception: a patch with Field Notes and any modulation cable can differ slightly between audio runs (about 0.1 dB in loudness in the case we measured).

SECTION 10

Documentation screenshots

Headless renders pictures of a patch, but the nodes themselves (their panels, knobs and dropdowns) can only be photographed in the real app. Infinite ships a screenshot tool for that, so the manuals always match the build. It launches the app, drives it over the local control port, and saves one PNG per node at 2x.

WANT	RUN
One node	`python3 scripts/node_screenshot.py "Wavetable"`
Every node type	`python3 scripts/screenshot_all_nodes.py --out docs/node_screenshots --settle-seconds 0.6`

Both need a Mac that is unlocked and awake, because the app must have a real window; they write the control token to `~/Library/Application Support/Infinite/control_token`. The Node Field Guide pulls its pictures from `docs/node_screenshots/`, and the brand screenshot standard (2x, real values, no personal paths) is in `docs/brand/BRAND.md` section 10.1.

SECTION 11

Where to go next

WANT	LOOK AT
Every node and its controls	`Infinite --describe`, and the Node Reference Manual PDF
The exact file format	`docs/reference/patch-format.md`
What can be wired to what	`docs/reference/connection-rules.md`
Starter patches	`assets/examples/authoring/`
Field (the expression language)	The Field language skill documents under `.claude/skills/field-language/`